Int J Biometeorol (2005) 49: 403–416
DOI 10.1007/s00484-005-0252-0

ORIGINAL ARTICLE

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Evapotranspiration under advective conditions

Received: 15 January 2004 / Revised: 11 November 2004 / Accepted: 26 November 2004 / Published online: 13 January 2005
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Abstract Arid and semi-arid regions are heterogeneous landscapes in which irrigated fields are surrounded by arid areas. The advection of sensible heat flux from dry surfaces is a significant source of energy that has to be taken into consideration when evaluating the evaporation from crops growing in these areas. The basic requirement of most of the common methods for estimating evapotranspiration [Bowen ratio, aerodynamic and Penman–Monteith (PM) equation] is that the horizontal fluxes of sensible and latent heat are negligible when compared to the corresponding vertical fluxes. We carried out measurements above an irrigated tomato field in a desert area. Latent and sensible heat fluxes were measured using a four-level Bowen machine with aspirated psychrometers. Our results indicate that under advective conditions only measurements carried out in the lowest layer are satisfactory for the estimation of latent heat fluxes and that the use of the PM equation with an appropriately parameterized canopy resistance may be preferable.

Keywords Evapotranspiration · Advection · Evapotranspiration models

This work was carried out by the author during her sojourn at the Wyler Department for Drayland Agriculture, Jacob Blaustein Institute for Desert Research, Ben Gurion University of the Negev, Israel.

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Introduction

Evaporation of water from vegetated surfaces is a key issue in the hydrology of arid and semi-arid areas. Sensible heat advection (downwind transport of energy in the horizontal plane) usually plays an important role in the evaporation process as it increases the amount of energy available for evaporation as well as the vapour pressure gradient between the advected parcel of air and the cooler moist vegetative surface (Prueger et al. 1996). Local advection is often associated with heterogeneous landscapes in which agricultural or naturally vegetated fields and dry non-vegetated areas are contiguous. A peculiar characteristic of arid and semi-arid landscapes is that the density of irrigated farmland is very low, as a result of which the irrigated fields are widely scattered within a dry matrix and resemble oases in a desert. The direct implication of this distribution is that advection will probably always play an important role in determining the evapotranspiration (ET) from irrigated fields in arid areas.

Baldocchi and Rao (1995) found that under advective conditions the latent and the sensible heat fluxes become invariant with downwind distance from the edge of the irrigated field after the fetch to height ratio exceeded 75:1. When latent and sensible heat fluxes were measured upwind of this threshold, a significant humidity deficit occurred, causing a vertical flux divergence of sensible and latent heat.

Kroon and Bink (1996) investigated the turbulent fluctuations in the internal boundary layer which forms in the wake of a dry to wet surface transition. They defined three periods with different thermal stability regimes: (1) morning with an unstable stratification up- and downstream of the transition, (2) afternoon when downstream of the transition the stratification changes from unstable to stable while

upstream the conditions remain unstable, (3) evening with stable stratification up- and downstream.

Eddy diffusivities for sensible and latent heat are not equal in the external part of the internal boundary layer where fluxes and gradients of scalars are known to be influenced by both up- and downwind sources. Near the ground, however, the horizontal gradients and their associated fluxes must both approach zero and the locally produced fluxes and gradients predominate (McNaughton and Laubach 1998). That is, an internal adjusted layer, which is in dynamic equilibrium with the new surface, is formed, the fluxes conform to the Monin–Obukhov similarity theory and the diffusivities are similar (Garratt 1990). This assumption is widely used in numerical modelling in order to define the lower boundary conditions within the atmosphere (e.g. Rao et al. 1974).

McNaughton and Laubach (1998) investigated the internal boundary layer under advection conditions and concluded that for strongly stable conditions the Bowen ratio (β_g) is <-1, and the relative errors in the fluxes become significant. That is, Bowen ratio measurements of the surface energy fluxes will usually be satisfactory, except during quite intense inversions. McNaughton and Laubach (1998) state as well that instruments should be located close enough to the ground to avoid errors that result from flux divergence. On the other hand, in the immediate vicinity of a canopy, horizontal homogeneity is not guaranteed as it depends on canopy characteristics and wind properties, and the one-dimensionality of fluxes can thus not be assumed.

The routine estimation of ET in arid zones is of great interest to irrigation engineers, plant physiologists, hydrologists and meteorologists. Lysimeters give a direct estimation of ET. However, they are expensive to build and maintain and therefore usually not available. Moreover, the variability of ET with lysimeters located in the same field reached 15% in some cases (Todd et al. 2000).

The three most common micrometeorological methods used to estimate actual ET above a crop are based on the Bowen ratio, the aerodynamic equation and the Penman–Monteith (PM) equation (Brutsaert 1982). The application of these methods requires horizontal advection to be negligible when compared to the magnitude of the vertical fluxes. In this case, the corresponding energy balance equation for an imaginary plane located above the canopy is:

$$H_a + E_a + R_N + G = 0 \quad (1)$$

where H_a and E_a are the vertical fluxes of sensible and latent heat, R_N is the net radiation and G is the soil heat flux all expressed in W m^{-2}. In temperate regions, the advection is usually negligible due to the relatively small differences in surface temperatures between the various fields and the size of the cultivated plots. As mentioned previously, this is not the case in arid zones.

The objective of our experimental work was to evaluate the height interval at which the abovementioned conventional methods used to estimate latent heat flux can be applied with confidence in an environment in which the advection of sensible heat may play a major role.

Materials and methods

The measurements were carried out between 25 September 1999 and 2 October 1999 in a commercial tomato field of about 330 m×220 m at Kibbutz Mashabei-Sade, in Israel (32°N, 34°E, elevation 300 m). To the west and south fallow land bordered the experimental field; to the north and to the east were fields planted with jojoba and olives (Fig. 1).

The crop was planted in rows 2 m apart. Data presented in this paper are for 30 September (day 273) and 1 October (day 274), 1 and 2 days after irrigation with brackish water, respectively. The canopy was well developed and reached a height of 0.44±0.08 m, and almost completely covered the ground, being the distance between the outer edges of two adjacent canopy rows of 0.09±0.11 m. The stage of tomato in this period is immature fruit, covering almost completely the surface.

The following measurements were made at four levels (0.8, 1.3, 1.8 and 2.3 m) above the soil surface: air and humidity using aspirated thermocouple psychrometers mounted on an automatic reversible system, and mean wind speed with cup anemometers (014A, Met-One) in another mast. The anemometer mast was placed north of the psychometric tower at 4 m distance; in this way the wind profiles were not perturbed when the wind was from the west-northwest and north-northwest, and the relation fetch/height was satisfactory. The anemometers were calibrated relative to each other in the field and to one of them in a wind tunnel.

The turbulence created near the top of the canopy is avoided when the measurements are taken in the inertial sublayer. Conventionally, the inertial sublayer is above a level referred to as z^* of the order of 2 to a few times the canopy height (h), and the roughness sublayer is below this level. We made the first measurement at 0.8 m—it represents about twice the canopy height. With more precision, z^* is defined by Cellier and Brunet (1992) as 3–4 times the inter-row spacing (δ) above displacement height (d=0.75h); we have a δ=0.09 m due to the canopy covering almost completely the ground in the immature stage, and then z^* is 0.7 m. The tomato plants are not tutored.

A four-level Bowen machine with aspirated thermocouple psychrometers was used, obtaining the dry and wet bulb temperature. The position of adjacent psychrometers was exchanged every 5 min during the 30-min integration period. The psychrometers were lined with polyurethane boards and covered by aluminum foil both on the inside and on the outside. Wind speed in the psychrometer duct close to the sensors was estimated at 4 m s^{-1}. Copper-constantan thermocouples were used as temperature sensors. Sensors from adjacent heights were measured differentially with a CR23X data logger. The paired thermocouples were tested in the laboratory by measuring temperature differences when introduced into a well-stirred water bath. The pairs

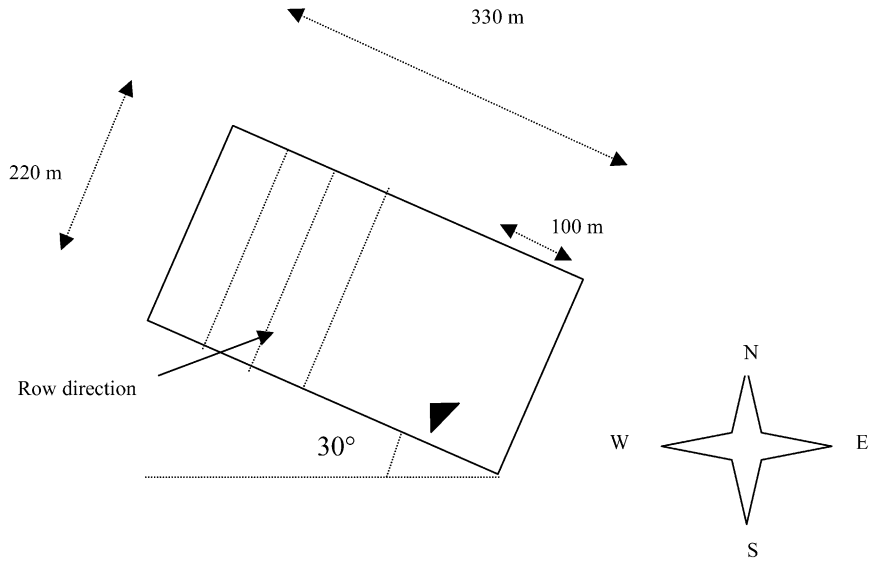

Fig. 1 Layout of experimental field, ◄ location of the instruments

of psychrometers were tested in the laboratory and in the field. In both cases, the intakes of both psychrometers were introduced into a box with a hole in the distal end, thus ensuring that the temperature of the air reaching the sensors was similar. Typical differences that ranged from 0.02°C to 0.2°C for dry and wet bulbs, respectively, were measured. Particular attention was paid to the proper functioning of the wet bulb. The wick was wetted using a Mariotte bottle located inside the psychrometer downwind of the sensors, thus practically eliminating the dangers of radiative heating of the water reservoir. The water supply system functioned satisfactorily even during events of very high temperature and unusually low relative humidity (35°C and 25%, respectively).

Net radiation was measured 2 m above the canopy (Q-7, Campbell Scientific) and soil heat flux was measured with two heat flux plates (HFT-3, Campbell Scientific) buried at a depth of 2 cm in the soil, one midway between the rows and the other one close to the row under the canopy. The distance between the outer edges of two adjacent canopy rows was δ=0.09 m, then the ground surface was completely shaded. The output of all meteorological sensors was recorded digitally every 10 s with a Campbell CR23X data logger.

The predominant wind direction was from the northwest. Gusts from the west or northeast wind occurred usually before midday when the average wind speed was low. The wind speed increased and blew across the rows after 1300 hours. During the morning, profiles were usually weakly unstable, almost isothermal at noon and showed a weak inversion thereafter. The shapes of the profiles of temperature and water pressure were similar for both days, and those for day 274 are presented in Fig. 2.

The daily maximum temperatures were 35 and 33.8°C, and the minimum daily relative humidity 30 and 45% for days 273 and 274, respectively, as measured at a weather station located close by.

Throughout the 2-day period, neutral conditions prevailed and wind speed exceeded 3 m s^{-1} only during one 30-min interval. The aerodynamic parameters of the crop were obtained by applying the method of Robinson (1962) to this time interval and d=0.338 m and z_o=0.0151 m were obtained. In terms of the height of the obstacles (h), $z_o/h \approx 0.03$, and $d/h \approx 0.77$ which compare well with those reported by Brutsaert (1982).

In Fig. 3, the evolution of the Richardson numbers for each layer is presented for both days. Stable atmospheric conditions were observed after 1330 hours for the uppermost layers while at the same time the conditions in the lowest layer remained close to neutral.

We concentrate on the profiles measured after 1400 hours when stable atmospheric conditions prevailed and the evaporation was highest because of dry hot air blowing across the rows.

The top of the internal boundary layer during this period, with wind blowing from the west-northwest and north-northwest, was estimated as being at heights of 2.6 and 3.4 m, respectively (Kaimal and Finnigan 1994).

Flux computation

The aerodynamic method

Within the inertial sub-layer the fluxes are assumed to be constant with height (or not to vary by >10%), and the so called aerodynamic method may be used for the computation of latent and sensible heat fluxes provided that the profile measurements are carried out in a height interval that is 0.01 of the fetch. In order to comply with this requisite, we used only the data from 1400 to 1830 hours, during which the fetch was at least 230 m and therefore all the measurements reported were carried out within the internal sub-layer. In order to minimize instrumentation error we carried out computations only during those intervals

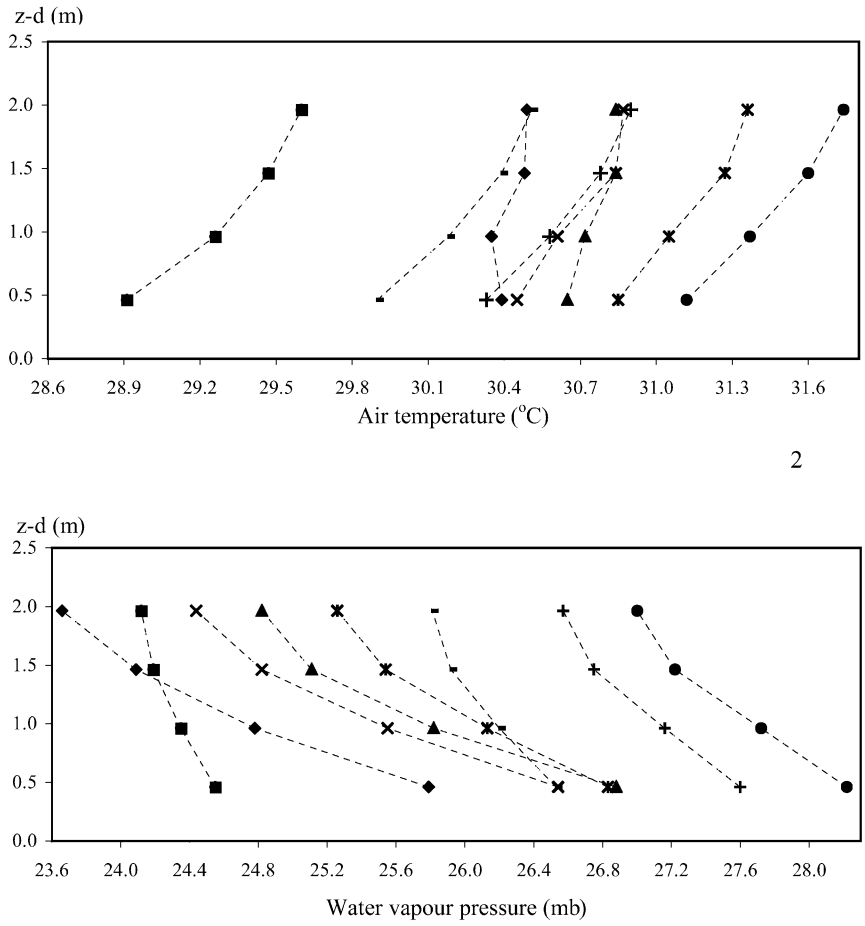

Fig. 2 Profile of air temperature (*upper panel*) and water vapour pressure (*lower panel*) for day 274 are shown, for ♦ 1400 hours, ▲ 1430 hours, × 1500 hours, * 1530 hours, ● 1600 hours, + 1630 hours, - 1700 hours, and ■ 1730 hours

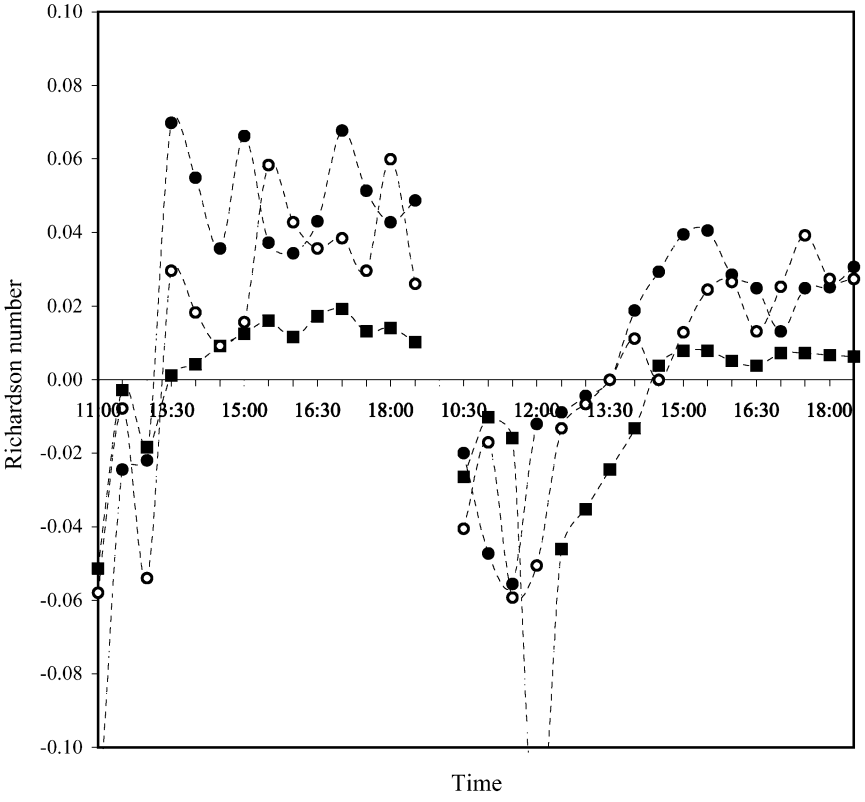

Fig. 3 Daily course of the Richardson numbers; data for day 273 are given on the *right*, and for day 274 on the *left*. ■ $z_j - d$=0.67 m, ● 1.19 m, ○ 1.69 m

during which the wind speed at the lowest layer exceeded 2.0 m s^{-1}.

The aerodynamic latent and sensible heat fluxes are computed as:

$$\mathrm{ET}_j^{\mathrm{Ae}} = -\rho L_{\mathrm{v}} u_{*_j} q_{*_j}; \quad H_j^{\mathrm{Ae}} = -\rho C_{\mathrm{p}} u_{*_j} T_{*_j} \tag{2}$$

where ρ is the air density and C_{p} is the air specific heat; u_{*j}, T_{*j} and q_{*j} are scaling parameters for wind speed, temperature and specific humidity, respectively. j is the layer, defining the mean geometrical heights as: $z_j - d = \sqrt{(z_{i+1} - d)(z_i - d)}$, between the levels i: 0.8, 1.3, 1.8 m and 2.3 m, define the layers: $z_j - d = 0.67$ m, 1.19 m and 1.69 m and are denoted by the subscripts L, I and U for lower, intermediate and upper layers, respectively.

H_j^{Ae} and $\mathrm{ET}_j^{\mathrm{Ae}}$ were evaluated at three height intervals for both days using (Businger et al. 1971; Dyer 1974; Kaimal and Finnigan 1994):

$$F_{*_j} = \frac{k\,(F_{i+1} - F_i)}{\ln\left(\frac{z_{i+1}-d}{z_i-d}\right) - \psi_{\mathrm{f}}\left(\frac{z_{i+1}-d}{L_j}\right) + \psi_f\left(\frac{z_i-d}{L_j}\right)} \tag{3}$$

where F is the variable (u, T or q) and i are the levels of the observed value; and ψ_{m}, ψ_{h} and ψ_{v} are the stability correction terms for momentum, temperature and specific humidity, respectively. For stable atmospheric conditions $\psi_{\mathrm{m}}=\psi_{\mathrm{h}}=\psi_{\mathrm{v}}$ and is computed as:

$$\psi = -5\frac{z_j - d}{L_j} \tag{4}$$

with the Monin–Obukhov length (L_j,) defined as:

$$L_j = (z_j - d)\frac{(1 - 5\mathrm{Ri}_j)}{\mathrm{Ri}_j}; \quad \text{with}\,\mathrm{Ri}_j > 0 \tag{5}$$

where Ri_j is the gradient Richardson number for layer j.

We use the mean absolute error (MAE):

$$\mathrm{MAE} = \frac{\sum_{\mathrm{i}=1}^{\mathrm{N}} |P_i - O_i|}{N} \tag{6}$$

as a criterion for estimating the differences between observed (O_i) and predicted (P_i) values.

Bowen ratio

The Bowen ratio is defined as:

$$\beta o_j = \gamma\frac{k_{\mathrm{H}}\,(T_{i+1} - T_i)}{k_{\mathrm{v}}\,(e_{i+1} - e_i)} \tag{7}$$

where T_i (°C) and e_i (mbar) are temperature and vapour pressure at level i , γ (mbar °C^{-1}) is the psychrometric constant, and j is the layer.

The main assumption in the derivation of the Bowen ratio is that the turbulent transfer coefficients for heat and water vapour are identical (k_{h}=k_{v}) or alternatively that the ratio is constant ($k_{\mathrm{h}}/k_{\mathrm{v}} = \alpha$).

Bink (1996) stated that the relation $k_{\mathrm{h}}/k_{\mathrm{v}}$ is approximately equal to 1 when the ratio of height to fetch, z/x, is $\leq$0.005. Our values for the three mean geometrical heights (0.67, 1.19 and 1.69 m) were 0.003, 0.005 and 0.007, respectively. We computed the Bowen ratio for these layers (βo_j) assuming k_{h}=k_{v}. The cases for which $-1.3 < \beta o_j < -0.75$ (sunset or sunrise) were not considered (Ortega-Farias et al. 1996).

Applying error analysis to β_o, its error ($\delta\beta_o$) is given by:

$$\delta\beta_{o_j} = \left|\frac{\delta\beta_{o_j}}{\delta\Delta T}\right|\delta\Delta T + \left|\frac{\delta\beta_{o_j}}{\delta\Delta e}\right|\delta\Delta e \tag{8}$$

where $\Delta e = e_{i+1} - e_i$ and $\Delta T = T_{i+1} - T_i$ corresponding to β_o of the layer j

$$\delta\beta_{o_j} = \left|\frac{\gamma}{\Delta e}\right|\delta\,(\Delta T) + \left|-\gamma\frac{\Delta T}{(\Delta e)^2}\right|\delta\,(\Delta e) \tag{9}$$

$$\delta\beta_{o_j} = \beta_{o_j}\left[\frac{\delta\,(\Delta T)}{\Delta T} + \frac{\delta\,(\Delta e)}{\Delta e}\right] \tag{10}$$

$\delta\beta_o$ depends on the resolution limits of the sensors and is inversely proportional to Δe and ΔT. In this experiment the air temperature is measured with thermocouples that give errors of about 0.02°C and a vapour pressure about 0.2 mbar; therefore a resolution for $\delta(\Delta T)$ is 0.04°C and for $\delta(\Delta e)$ is 0.4 mbar. The latent heat flux can be estimated through the Bowen ratio as:

$$\mathrm{ET}_j^{Bo} = \frac{(R_N - G)}{1 + \beta_{o_j}} \tag{11}$$

where R_{N} is the net radiation and G is the soil heat flux in W m^{-2}. The ET, ET_j^{Bo}, of each layer was obtained applying Eq. (11).

Verma et al. (1978) showed that the relation $k_{\mathrm{h}}/k_{\mathrm{v}}$ is 1 in neutral conditions, and increases under strong stable conditions ($z/L > 0.02$). Lang et al. (1983), however, postulated that the ratio diminishes with increasing stability. Blad and Rosenberg (1974) on the other hand concluded that when $-2.5<\mathrm{Ri}<0.025$, the relationship $k_{\mathrm{h}}/k_{\mathrm{v}}$ is close to 1 and the error in the Bowen estimation is <10%. The Richardson numbers for the lowest layer (Fig. 3) do not exceed 0.02 during both days, and it is therefore safe to assume that k_{h}=k_{v}.

After applying the error analysis to Eq. (11) the relative uncertainty in ET^{Bo} is given by:

$$\frac{\delta\mathrm{ET}}{\mathrm{ET}_j^{Bo}} = \frac{\delta R_n + \delta G}{|R_n - G|} + \frac{\delta\beta_{o_j}}{|1 + \beta_{o_j}|} \tag{12}$$

A typical accuracy of R_n is 5% and an average uncertainty in G is 5%. The first term of the equation gives the biggest errors in the evening and the second term will be analysed latter.

PM method

The ET may be estimated from the PM equation:

$$\mathrm{ET_{PM}} = \frac{s(R_n - G) + \rho C_\mathrm{p}[e_\mathrm{s}(T_\mathrm{a}) - e]/r_\mathrm{a}}{s + \gamma[1 + (r_\mathrm{c}/r_\mathrm{a})]} \tag{13}$$

where s (mbar °C^{-1}) is the slope of the saturation-vapour pressure versus the temperature curve, γ (mbar °C^{-1}) is the psychrometric constant, $e_\mathrm{s}(T)$ (mbar) is the saturated vapour pressure at air temperature T_a (°C), e(mbar) is the vapour pressure, r_a (s m^{-1}) the aerodynamic "bulk" resistance to sensible heat transfer of the air and r_c (s m^{-1}) the canopy resistance. The dependence of the latter on some easily measured environmental parameters like soil water content has been proposed (Rana et al. 1997, 1998) and it would allow the computation of actual latent heat fluxes using measurements of the relevant quantities at one height only.

r_a (s m^{-1}) is defined by:

$$r_\mathrm{a} = \frac{[\ln(z_\mathrm{r} - d)/z_\mathrm{o}]^2}{k^2 u_\mathrm{r}} \tag{14}$$

where z_r is the level at which measurements were carried out, u_r (m s^{-1}) is the wind speed at height z_r and k is the von Karman constant. The aerodynamic resistance of sensible heat transfer is assumed to be equal to the resistance for momentum because under neutral stability the diffusion coefficient for turbulent transfer in air for heat, moment and water vapour are equivalent (Monteith and Unsworth 1990). Stability corrections to Eq. 14 can be introduced (Thom and Oliver 1977) but the equation becomes cumbersome. Close to the canopy the mechanically generated turbulence predominates and stability corrections are probably minor. Kroon and de Bruin (1993) proposed computing a canopy resistance when advective conditions prevail, using the PM method.

Results

Aerodynamic method

In the second column of Table 1, the relationship between $(H_j^\mathrm{Ae} + \mathrm{ET}_j^\mathrm{Ae})$ and $R_\mathrm{N}-G$ is presented. For the lowest layer, the slope is 0.98 and the MAE is 52.1 W m^{-2}, while for the other layers, these relationships are not as good. The intercept of the linear least-squares regression is significantly different from zero (level of significance α=0.05) for the 0.67 m layer, suggesting an amount of measurement uncertainty. The conventional approach is that changes in fluxes of up to 10% within the constant stress layer are admissible. Our results indicate that the vertical fluxes above 0.67 m exceed this value and that flux divergence in these layers cannot be ignored.

Bowen ratio

The error $\delta\beta_{o_j}$ is shown in Fig. 4 (Eq. 10), the less uncertain β_{o_j} refers to the upper layer where the ΔT and Δe are smaller than in the lower layer where great ET occurs. The errors between the lower and intermediate layer are of similar magnitude. The biggest error is observed at about 1800 hours and 1830 hours. At these times β_{o_j} was derived according the criteria of Ortega-Farias et al. (1996). Later, we will discuss how these errors would affect the latent heat fluxes.

The temperature inversion (Fig. 2) leads to the transport of the sensible heat flux downward after 1400 hours. However, from 1400 until 1530 hours, ΔT is smaller than

Table 1 Linear regression statistics. R^2 Determination coefficient, *MAE* mean absolute error

Mean geometric height of layer[a]	$H_j^\mathrm{Ae} + \mathrm{ET}_j^\mathrm{Ae}$[b]	βo_j[c]	$\mathrm{ET}_j^\mathrm{Bod}$
0.67 m (Lower)	0.98 ($R_\mathrm{N}-G$)−39.8	–	1.16 $\mathrm{ET}_L^\mathrm{Ae}$+1.90
	$R^2 = 0.96$	–	$R^2 = 0.89$
	MAE = 52 W m^{-2}	–	MAE = 50 W m^{-2}
1.19 m (Intermediate)	0.62 ($R_\mathrm{N}-G$)+8.4	0.61 βo_L−0.11	1.10 $\mathrm{ET}_L^\mathrm{Ae}$+39.3
	$R^2 = 0.86$	$R^2 = 0.97$	$R^2 = 0.78$
	MAE = 77 W m^{-2}	MAE = 0.06	MAE = 76 W m^{-2}
1.69 m (Upper)	0.80 ($R_\mathrm{N}-G$)−41.5	1.89 βo_L+0.13	1.22 $\mathrm{ET}_L^\mathrm{Ae}$−16.9
	$R^2 = 0.94$	$R^2 = 0.85$	$R^2 = 0.78$
	MAE = 87 W m^{-2}	MAE = 0.07	MAE = 66 W m^{-2}

[a]The geometric mean of the three height intervals considered

[b]The statistics of the linear regression between $R_\mathrm{N}-G$ and the fluxes $(H_j^\mathrm{Ae} + \mathrm{ET}_j^{Ae})$ computed using the aerodynamic approach

[c]The statistics of the correlation of the Bowen ratios with the Bowen ratio of the lowest layer (βo_L)

[d]The statistics corresponding to the linear regression between latent heat fluxes (computed via the Bowen ratio) to the latent heat flux of the lowest layer computed using the aerodynamic approach $(\mathrm{ET}_L^\mathrm{Ae})$

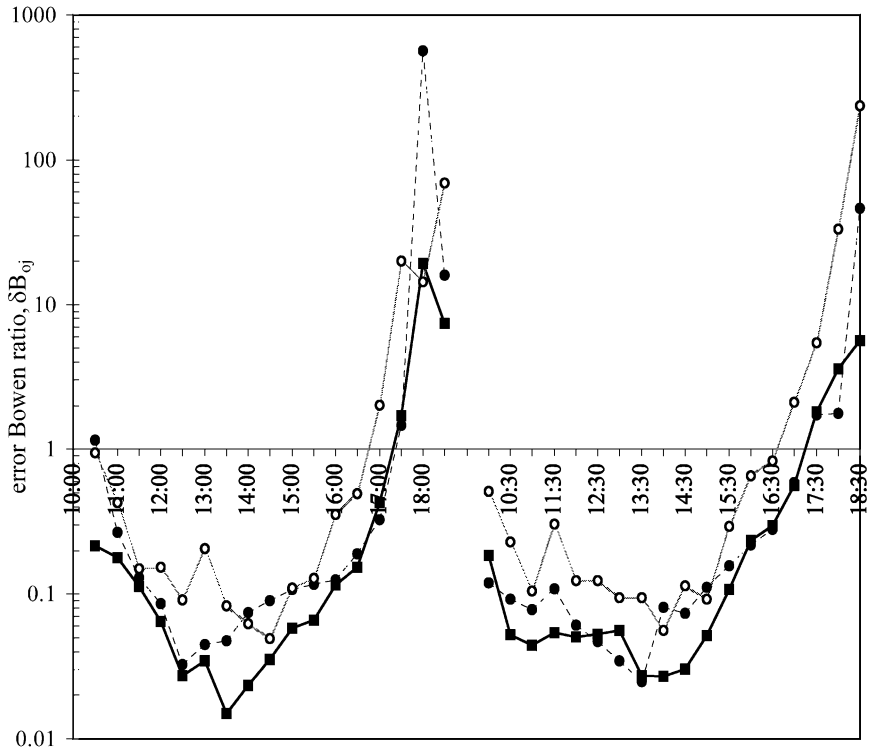

Fig. 4 The error of the Bowen ratio, $\delta\beta_{o_j}$, for the layers $z_j - d$=0.67 m (■), 1.19 m (●), and 1.69 m (○)

after 1530 hours when the advection is stronger. After 1530 hours, the temperature inversion is kept almost constant in the top and intermediate layers due to the downward contamination of warm air advection. While in the bottom layer the result is a strong latent cooling that maintains an oasis effect (the biggest ΔT).

The Δe decreased with the height in the dry–wet surface transition. But in the intermediate layer it was observed that Δe is close to that of the lower layer after 1530 hours; after this time strong winds blow and great velocity gradients in the bottom layer suggest a more effective transport of heat latent flux upwards. This kind of feature is observed in the changes of the Bowen ratio.

The evolution of β_{o_j} with height as a function of time for day 274 is presented in Fig. 5 (for day 273 the observed behaviour is similar). The pattern of change of the Bowen ratio with height is altered at 1530 hours (Fig. 5). The upper layer has a more negative value than the intermediate layer after 1530 hours. This trend reflects an increase in advected energy, which can be ascribed to an increase in the mean horizontal wind speed.

In Fig. 6 and in the third column of Table 1, the relationship between the Bowen ratios in each layer with that of the Bowen ratio at 0.67 m is presented. As is evident from this figure, the Bowen ratio at 0.67 m does not correlate well with the Bowen ratio of the other layers (1.19 and 1.69 m). The coefficients of correlation between them were high (Table 1), but the slope of the linear regression is significantly different from 1. The intercepts are, however, not significantly different from zero. In Fig. 6, an inflexion point near $\beta_{o_L} = -0.20$ can be observed and this corresponds to the change in the sign of profiles measured after 1530 hours, for both days. We note that the error in β_{o_j} at about 1530 hours is small, i.e. <10%, and the inflexion point is not due to an error of β_{o_j} (Fig. 4).

In the total internal boundary layer, according to McNaughton and Laubach (1998), $k_H < k_v$ when it is affected by advection. We divided this layer into three layers, where in the lower layer the stability was neutral (Fig. 2); under this condition $k_H = k_v$ is valid (Blad and Rosenberg 1974).

The slope between β_{o_L} (lower layer where $k_H = k_v$) and β_{o_I} (intermediate layer) differs from 1 (Table 1). From this we can deduce that in the intermediate layer $k_H \neq k_v$ during local advection (Fig. 6). The results presented here indicate that even though the Bowen ratio for the intermediate layer conforms to the height/fetch ($z/x \leq 0.005$) ratio, a requirement set by Bink (1996), it is different from that computed for the lowest layer, leading us to conclude that only in the latter can advective effects be neglected $((z-d)/h = 1.5)$. In advection, the temperature and vapour pressure profiles are determined for the diffusivity originating from the underlying surface and from the upwind area; in our study only the lower layer adjusted well to the surface layer. This illustrates that where the diffusion originates from different sources the simple gradient theory does not hold.

ET using the Bowen ratio

The fractional error of ET_j^{Bo} depends on $(R_N - G)$ and β_{o_j}; the second term of Eq. (12) is analysed for the three layers.

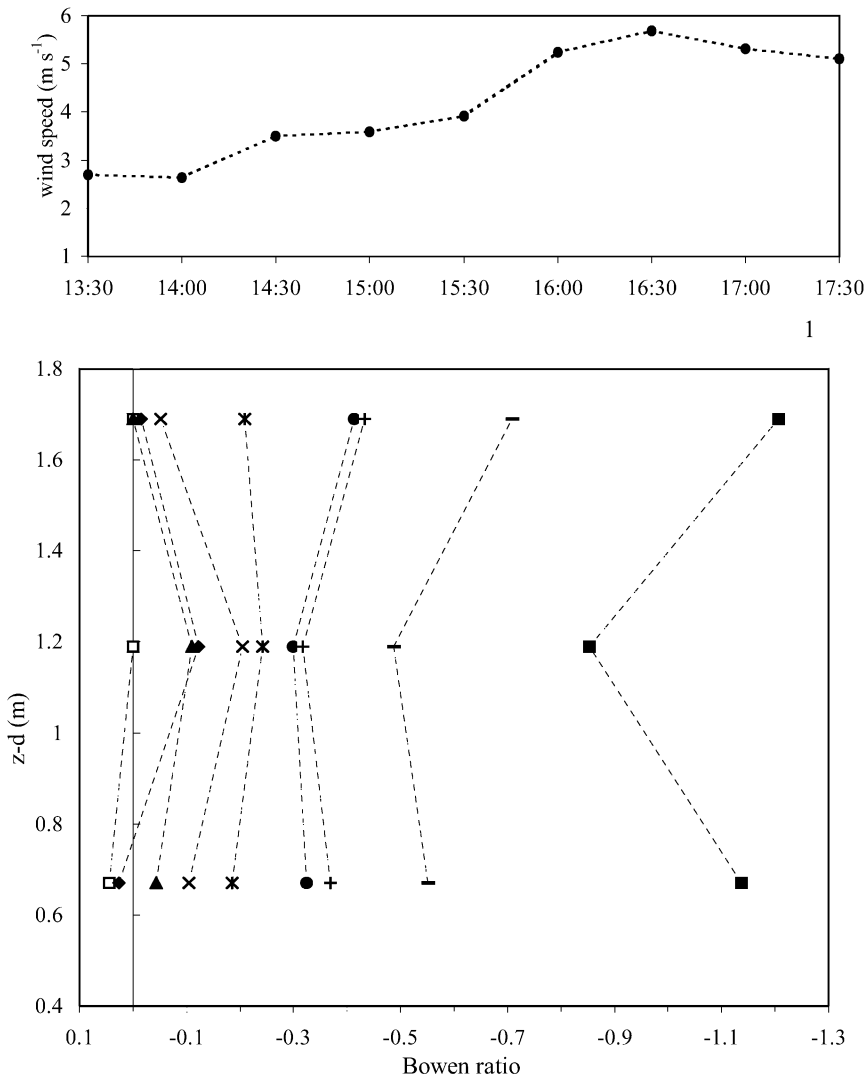

Fig. 5 The daily course of wind speed at 2.3 m on 1 October (day 274) (*upper panel*). Bowen ratio (*lower panel*) for layers $z_j - d$: 0.67 m, 1.69 m, and 1.19 m. □ 1330 hours, ♦ 1400 hours, ▲ 1430 hours, × 1500 hours, * 1530 hours, • 1600 hours, + 1630 hours, - 1700 hours, ■ 1730 hours

The relation $\delta\beta_{o_j}/|1+\beta_{o_j}|$ is shown in Fig. 7. Before 1600 hours this term contributed to <20% of the fractional error of ET_j^{Bo} although the highest is for the upper layer. On the other hand, this term is greater after 1700 h and we are less certain of the values obtained ET_j^{Bo}.

The fourth column in Table 1, shows the relation between ET calculated using the Bowen ratio, ET_j^{Bo} for each layer, and the ET of the lowest layer estimated via the aerodynamic method, $\mathrm{ET}_L^{\mathrm{Ae}}$. The best slope and intercept of the linear least-squares regression (slope closer to 1 and intercept closer to zero) are for the 0.67-m layer. This relationship deteriorates with height and this can be tentatively ascribed to the fact that probably $k_{\mathrm{h}} \neq k_{\mathrm{v}}$, as more stable conditions prevail at greater heights.

The fractional error in $\mathrm{ET}_L^{\mathrm{Ae}}$ (Eq. 2) is proportional to the errors $\delta\Delta u$ and $\delta\Delta q$ so that:

$$\frac{\delta E_L^{Ae}}{E_L^{\mathrm{Ae}}} = \frac{\delta\Delta u}{\Delta u_L} + \frac{\delta\Delta q}{\Delta q_L} \tag{15}$$

where $\Delta u = u_{i+1} - u_i$ and $\Delta q = q_{i+1} - qi$, $\delta\Delta u = 2\delta u$, $\delta\Delta q = 2\delta q$. δu and δq are the error in the wind speed and the specific humidity, respectively.

In this equation, the second term is more weighted than the first when Δq decreases towards the evening when it has the largest error. Both, $\mathrm{ET}_L^{\mathrm{Ae}}$ and ET_j^{Bo}, with the greatest fractional error toward the evening although the absolute value of ET is not large in these hours.

In Fig. 8, the fluxes of ET_L^{Bo} and H_L^{Bo} are presented (the data of 1730 hours were interpolated because the value of βo_L was between −0.75 and −1.3 m; Ortega-Farias et al. 1996). ET_L^{Bo} peaked at noon (500 W m^{-2}) while the H_L^{Bo} is negative after 1400 hours and its maximum value is about 170 W m^{-2} in the evening.

Advection conditions in our trial were not so intense as those reported by Prueger et al. (1996) and Todd et al. (2000). In the former case, a mean sensible heat flux in the order of 160 W m^{-2} was measured (higher than our average of 80 W m^{-2} for the same period), and in the latter case, the latent heat flux exceeded 900 W m^{-2}.

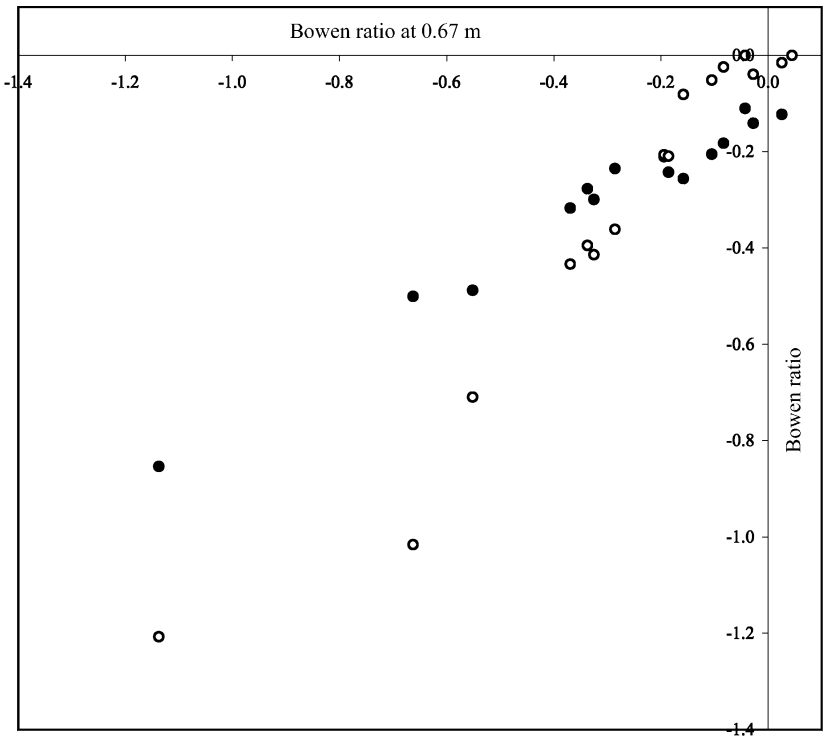

Fig. 6 The Bowen ratio (β_{o_j}), for the layers $z_j - d$: 1.19 m (•) and 1.69 m (○), as a function of the Bowen ratio at 0.67 m (β_{o_L}), for both days

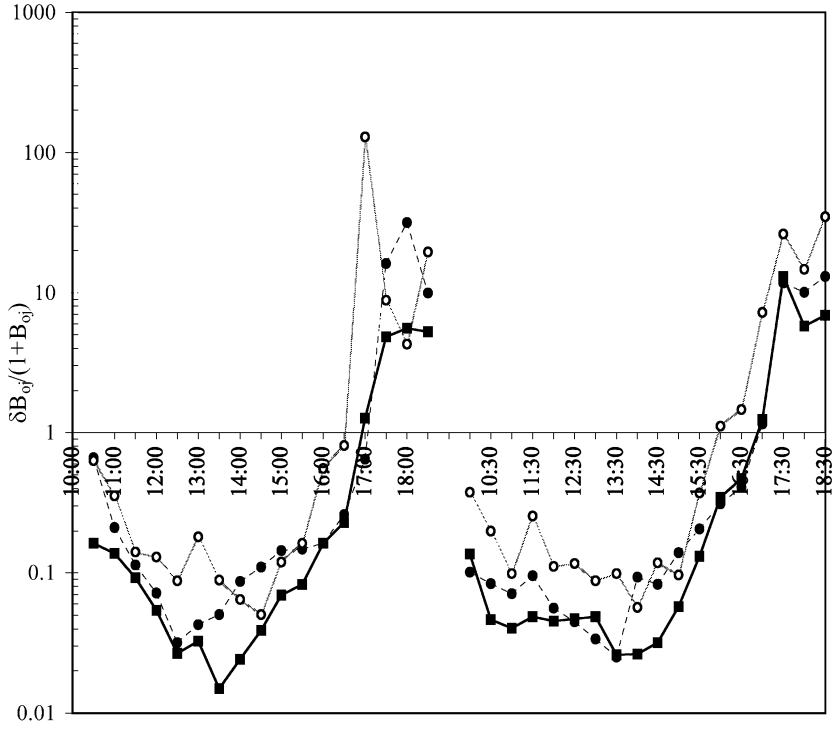

Fig. 7 The second term of Eq. 12, $\delta\beta_{o_j} / |1 + \beta_{o_j}|$, with time for the layers: $z_j - d$=0.67 m (■), 1.19 m (•), and 1.69 m (○)

Todd et al (2000) compared half-hourly values of latent heat fluxes, computed via the Bowen ratio method, to lysimeter data and indicated that the agreement was best (with the index of agreement close to 0.87) when the mean latent flux was <400 W m^{-2}. We can therefore conclude that for the conditions prevailing in our trial the error in the Bowen ratio estimate of latent flux is not large.

Potential ET using the PM approach

In order to predict latent heat fluxes using the PM equation the value of r_c has to be known. To evaluate the effect the advective conditions may have on the daily course of the bulk canopy resistance, we computed the latter by inverting the PM equation (Eq. 13), using ET_L^{Ae} instead of ET_{PM}.

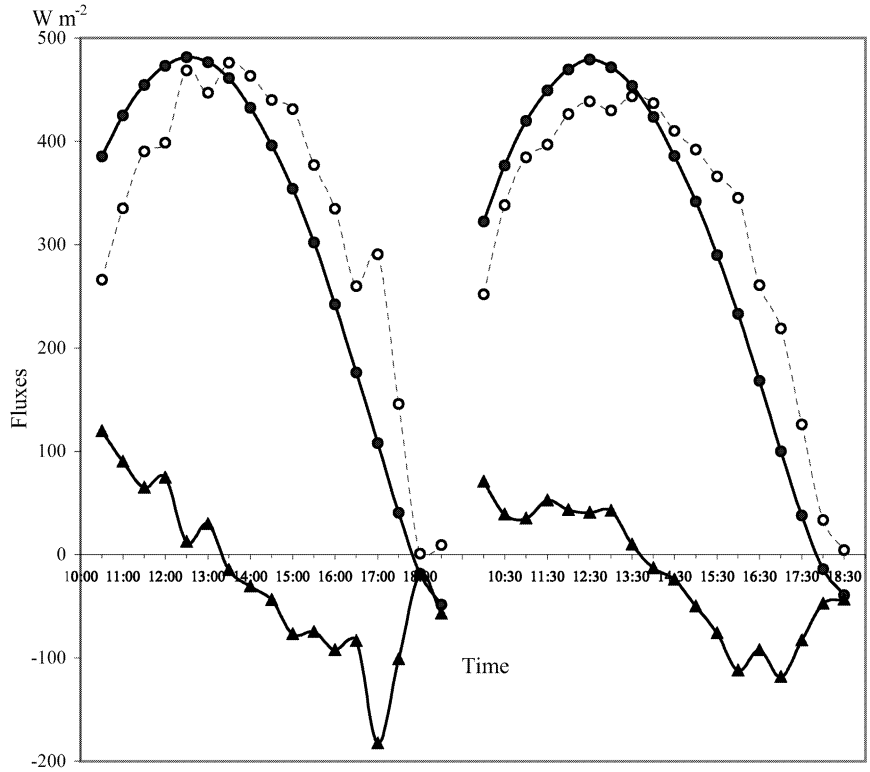

Fig. 8 Measured and computed fluxes for day 273 (*right*) and day 274 (*left*). The sensible heat flux (H_L^{Bo}) (▲) and the latent heat flux (ET_L^{Bo}) (○) were computed from the Bowen ratio at 0.67 m. • $R_\mathrm{N}-G$

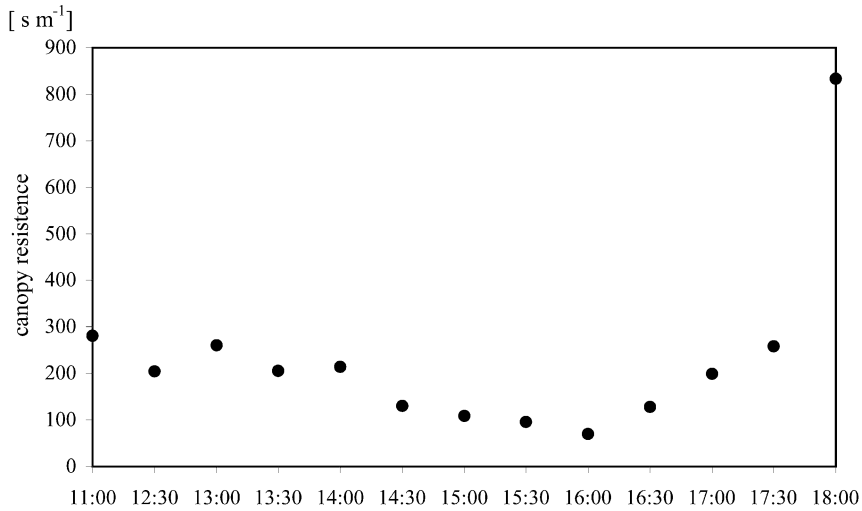

Fig. 9 The computed canopy resistance for day 273 (r_c) deduced by inverting the Penman–Monteith (PM) equation (Eq. 13)

The daily course of r_c computed for day 273 is presented in Fig. 9. The tendency of the computed resistances roughly conforms to the measurements of Papadakis et al. (1994), even though environmental conditions for both setups were very different. We may define our tomato crop as well watered and the derived crop resistances those typical of such a situation. In these circumstances, the obvious environmental choice for the parameterization of r_c is the net radiation, which we carried out using a second-degree polynomial.

This parameterization was used in order to compute ET using the PM equation for day 274, and correlate it to $\mathrm{ET}_L^{\mathrm{Ae}}$ (as we have established previously that the lowest level is our height of reference). The relation obtained for day 274 was: $\mathrm{ET}_\mathrm{PM} = 1.02\mathrm{ET}_L^{\mathrm{Ae}} + 8.07$ (R^2=0.66) (Fig. 10). The slope is not significantly different from 1 nor is the intercept different from zero (P=0.05).

We are interested in examining the degree to which the ET computed using Eq. 13 is affected, under advective conditions, by the height at which measurements are carried out.

The relative error of ET may be related to the relative error in the selected input parameters (p) (Figuerola and Mazzeo 1997):

$$\frac{\delta\,(\mathrm{ET})}{\mathrm{ET}} = K^* \frac{\delta p}{p} \tag{16}$$

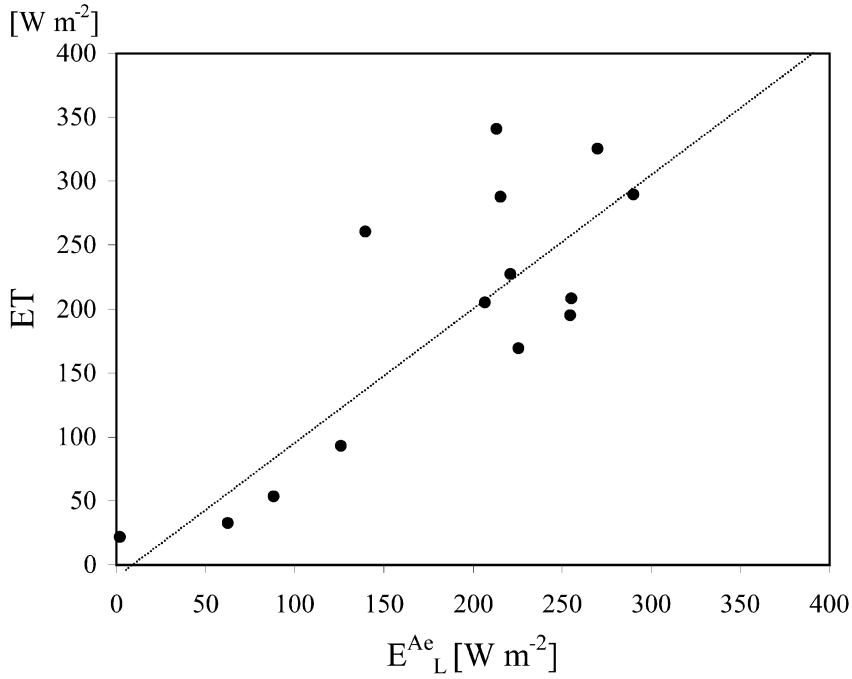

Fig. 10 A comparison of the latent heat flux from 1130 hours to 1830 hours for day 274 by using the aerodynamic method for 0.67 m (ET_L^{Ae}), and the PM equation for the estimation of evapotranspiration (ET_{PM}): using an approximation to the second-degree polynomial of the net radiation of day 273

and the coefficient K^* is obtained by applying propagation of error analysis.

The main sources of height-dependent errors in the computation ET_{PM} are the vapour pressure deficit and r_a. For the former the relative error is:

$$\left.\frac{\delta\,(\mathrm{ET})}{\mathrm{ET}}\right|_{\mathrm{De}} = \frac{\rho C_p \mathrm{De}/r_a}{[s(R_N - G) + \rho C_p De/r_a]}\frac{\delta(\mathrm{De})}{\mathrm{De}} = \frac{1}{[\frac{s(R_N-G)}{\rho C_p \mathrm{De}/r_a} + 1]}\frac{\delta(\mathrm{De})}{\mathrm{De}} = K_{\mathrm{De}}\frac{\delta(\mathrm{De})}{\mathrm{De}} \quad (17)$$

where $\mathrm{De}=[e_s(T_a)-e]$, and $\delta(\mathrm{De})/\mathrm{De}$ is the corresponding relative error.

K_{De} is always <1 provided R_N-G is positive. The average value of K_{De} is 0.6 (Fig. 11). The average values of $\delta(\mathrm{De})/\mathrm{De}$ for both measurement days are 0.05, 0.09 and 0.11 for 1, 1.5 and 2 m, respectively, yielding a relative error of 7% for the highest measurement level.

The relative error in ET due to a relative error in r_a is:

$$\left.\frac{\delta(\mathrm{ET})}{\mathrm{ET}}\right|_{r_a} = \left[\frac{-\rho C_p De/r_a}{s(R_N - G) + \rho C_p De/r_a} + \frac{\gamma(r_c/r_a)}{(s + \gamma(1 + (r_c/r_a)))}\right]\frac{\delta(r_a)}{r_a} = K_{r_a}\frac{\delta(r_a)}{r_a} \quad (18)$$

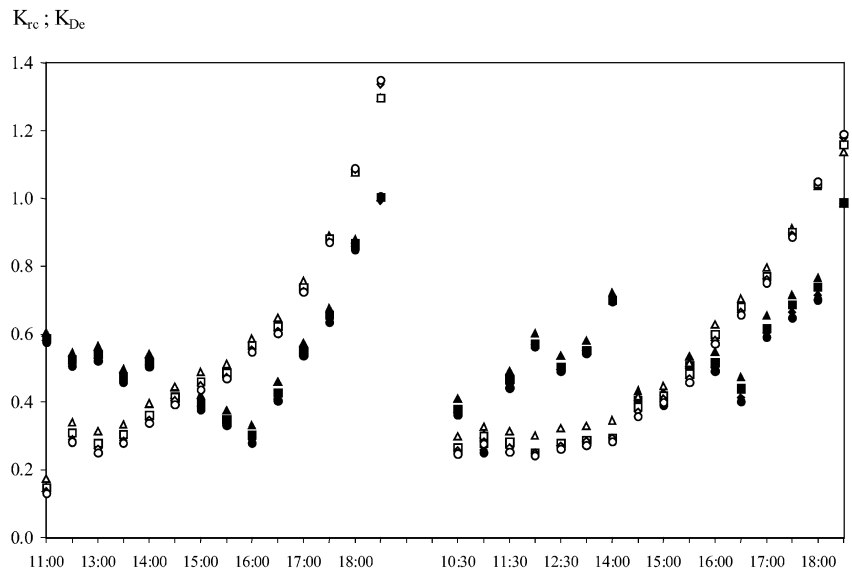

Fig. 11 The daily values of the coefficients K_{De} (*white symbols*) and K_{rc} (black symbols) for days 273 and 274 are shown. For the levels $z-d$: ▲ at 0.5 m, ■ at 1 m, ♦ at 1.5 m and ● at 2 m

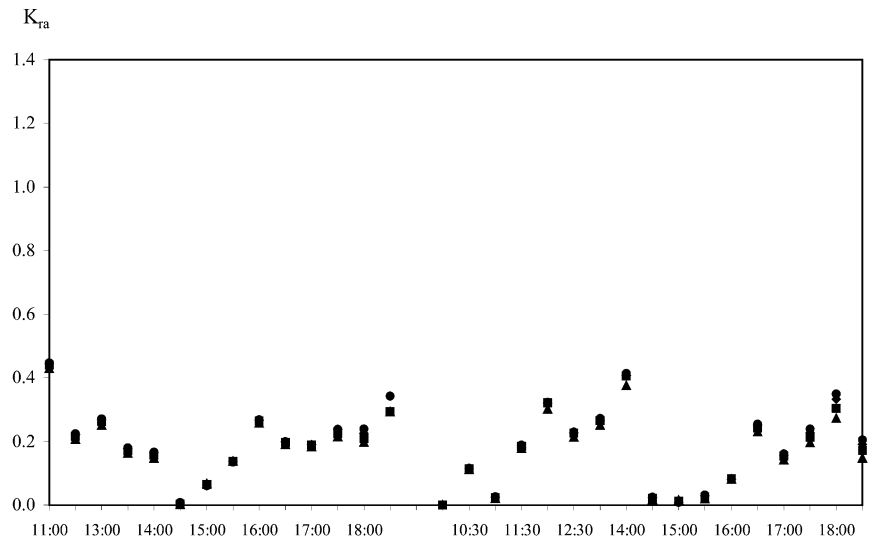

Fig. 12 The daily values of the coefficient K_{ra} for both days are shown. For the levels $z-d$: ▲ at 0.5 m, ■ at 1 m, ♦ at 1.5 m and at • 2 m

K_{r_a} was always <0.4 (Fig. 12). We define $\delta(r_a)/r_a$ as the relative error resulting from the exclusion of the stability correction in the computation of r_a, with

$$\delta(r_a) = r'_a - r_a$$

and

$$r'_a = \frac{\left(\ln\left(\frac{z_r-d}{z_o}\right) - \psi\left(\frac{z_r-d}{L}\right)\right)^2}{k^2 u_r} \tag{19}$$

Average values of $\delta(r_a)/r_a$ were 0.03, 0.11 and 0.07 for the three heights. The relative errors in $\delta(\mathrm{ET_{PM}})/\mathrm{ET_{PM}}$ increase therefore from 0.1 to 4%.

An additional source of error is the error in r_c. The relative error in $\mathrm{ET_{PM}}$ as a function of the relative error in r_c is:

$$\left.\frac{\delta(\mathrm{ET})}{\mathrm{ET}}\right|_{r_c} = \frac{\gamma(r_c/r_a)}{s+\gamma(1+(r_c/r_a))}\frac{\delta r_c}{r_c} = K_{r_c}\frac{\delta r_c}{r_c} \tag{20}$$

where K_{r_c} is always <1. If $(r_c/r_a) \sim 10$, as in our case, the average value of K_{r_c} is 0.6, and with average values of $\delta(r_c)/r_c$ of 0.09, 0.15 and 0.18 for the three height intervals of interest, the average errors in $\mathrm{ET_{PM}}$ are 5–10%.

The coefficients K_{De}, K_{r_c} and K_{r_a} have been assumed in the derivations carried out above to be invariant with height. Their diurnal change, for each of the three heights, is presented in Figs. 11 and 12, and it is evident that they are barely affected.

$\mathrm{ET_{PM}}$ is a function of the water vapour pressure, air temperature, wind speed, and net radiation and soil heat flux. The error in $\mathrm{ET_{PM}}$ due to errors in δe, δT_a, δu, and $\delta(R_n-G)$ are analysed below.

$$\left.\frac{\delta\mathrm{ET_{PM}}}{\mathrm{ET_{PM}}}\right|_e = \left[\frac{\rho C_p e/r_a}{s(R_N-G)+\rho C_p \mathrm{De}/r_a}\right]\frac{\delta e}{e} = \left[K_{\mathrm{De}}\frac{e}{\mathrm{De}}\right]\frac{\delta e}{e} = K_e\frac{\delta e}{e} \tag{21}$$

$$\left.\frac{\delta\mathrm{ET_{PM}}}{\mathrm{ET_{PM}}}\right|_{T_a} = \left[\frac{\rho C_p T_a s/r_a}{s(R_N-G)+\rho C_p \mathrm{De}/r_a}\right]\frac{\delta T_a}{T_a} = \left[K_{\mathrm{De}}\frac{sT_a}{\mathrm{De}}\right]\frac{\delta T_a}{T_a} = K_{T_a}\frac{\delta T_a}{T_a} \tag{22}$$

$$\left.\frac{\delta\mathrm{ET_{PM}}}{\mathrm{ET_{PM}}}\right|_u = \left[\frac{K_{r_a}}{r_a}\frac{\partial r_a}{\partial u}\right]\delta u = K_{r_a}\frac{\delta u}{u} = K_u\frac{\delta u}{u} \tag{23}$$

$$\left.\frac{\delta\mathrm{ET_{PM}}}{\mathrm{ET_{PM}}}\right|_{(R_n-G)} = \left[\frac{s(R_N-G)}{s(R_N-G)+\rho C_p \mathrm{De}/r_a}\right] \times \frac{(\delta R_N+\delta G)}{(R_N-G)} = K_{(R_N-G)}\frac{(\delta R_N+\delta G)}{(R_N-G)} \tag{24}$$

With δe=0.2 mb, δT_a=0.02°C, δu=0.11 m s^{-1}, and $\delta R_N+\delta G$=10%, we can discuss the error as follows:

$K_{\mathrm{De}}\sim$0.6, $(e/\mathrm{De})\sim$1.4 then $K_e\sim$0.8, and the fractional error is $(\delta e/e)\sim$0.01; consequently with respect to vapour pressure $\delta\mathrm{ET_{PM}}/\mathrm{ET_{PM}}|_e \sim 1\%$. The values of K_e are shown in Fig. 13; these appear to be more dispersed than those of K_{De} (Fig. 11) but follow the same trend.

$K_{r_a} \sim 0.2$ (Fig. 12) and $(\delta u/u)\sim$0.04 between the afternoon and evening, and an average value of $\delta\mathrm{ET_{PM}}/\mathrm{ET_{PM}}|_u \sim 1\%$; the largest error is at noon when the wind speed is low. $K_{\mathrm{De}}\sim$0.6, the factors $(s.T_a/\mathrm{De})\sim$4.3 and $(\delta T_a/T_a)\sim$0.001 result in $\delta\mathrm{ET_{PM}}/\mathrm{ET_{PM}}|_{T_a} \sim 0.2\%$; this was very small. K_{R_N-G} is 0.6 at midday and it decreases toward the evening to 0.1.

Measurement errors were certainly always present in the field data, but the idea that the upper layer is affected by the hot dry air from an upwind area is acceptable, while at the bottom layer the effect due to the vicinity of the canopy is avoided with a suitable closure of the balance energy equation. Moreover, if the wind speed is high

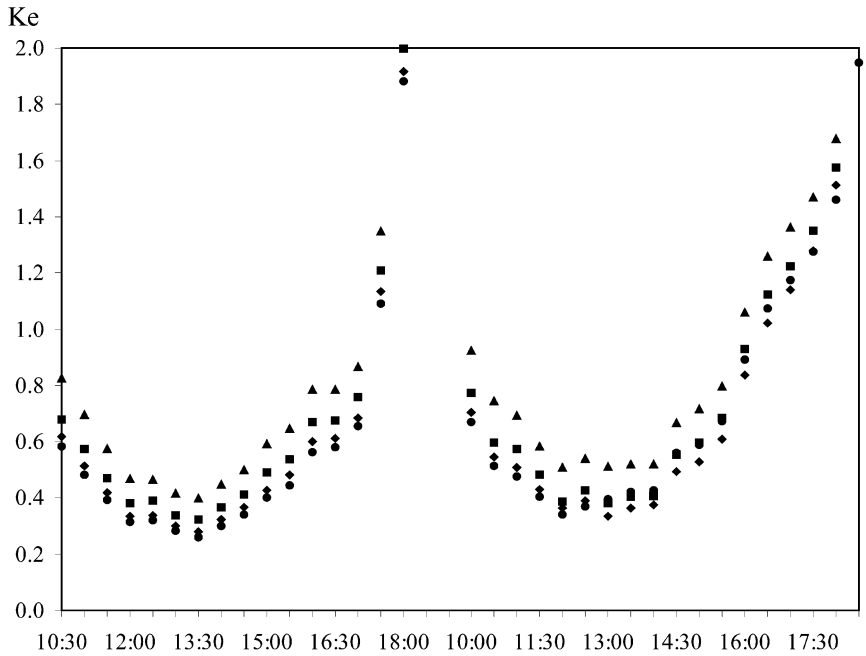

Fig. 13 The daily values of the coefficients K_e for days 273 and 274 are shown, for the levels $z-d$: ▲ at 0.5 m, ■ at 1 m, ♦ at 1.5 m and at ● 2 m

and the ground is shaded the heterogeneity of the canopy decreases. Polonio and Soler (2000) showed for winds >3 m s^{-1} the agreement between the Bowen ratio and the aerodynamic method to estimate the ET over a weakly heterogeneous canopy was acceptable. The comparison between both methods in our analysis (with local advection and high wind) of only the lower layer showed agreement. The more important error in the ET occurred in the evening when $\beta_{o_j} \approx -1$, having been eliminated from the compared values.

Conclusions

In this paper, the ET of a canopy under local advection conditions was investigated using the aerodynamic method, Bowen ratio and PM equation. The layers studied were $z_j-d=1.5h$, $2.7h$ and $3.8h$ (where h is the height of the canopy), and the upper boundary of the internal boundary layer was about $5h$. The local advection occurred after 1400 hours, but was not as extreme as that reported by Prueger et al. (1996) or Todd et al. (2000).

The Bowen ratio was not constant with height within the internal boundary layer. In the lowest layer ($z-d=1.5h$), the neutral atmospheric stability conditions prevail. In the upper layers ($z_j-d=2.7h$ and $3.8h$), stable conditions prevail and the lack of constancy with height may be ascribed to the difference in the turbulent transfer coefficients for heat and water vapour ($k_h \neq k_v$). In the lowest layer which was close to neutrality, ET_L^{Bo} matches ET_L^{Ae} and we may therefore conclude that $k_h=k_v$.

The lack of closure of the energy balance equation for the upper layers, even though these layers are well within the internal boundary layer, are the result of the effects of advection. Our results show that in conditions of local heat advection, as a result of an upwind dry-to-wet transition, the conditions in the lower layer ($z-d=1.5h$) remained close to neutrality and allow the computation of fluxes for sensible and latent heat using the common assumptions of lack of divergence and equality of transport coefficients for latent and sensible heat fluxes.

The bulk stomatal resistance was parameterized as a function of net radiation for the first measurement day. This parameterization was used in order to estimate the ET for the following day and good agreement between the thus calculated values and those computed using the aerodynamic equation for the lowest layer was obtained.

A sensitivity analysis showed that the largest errors in the estimation of ET_{PM} were due to possible errors in the estimation of the vapour pressure deficit and the canopy resistance. Minimizing the former, by setting up the instruments as close as possible to the top of the canopy, indicated that the errors in the PM equation used to estimate ET derive mainly from errors in the estimation of r_c. If the latter can be satisfactorily parameterized, as was the case in this study, it appears that it is advantageous to use the PM equation to estimate actual ET under advection.

Acknowledgements During her visit to the Jacob Blaustein Institute of the Ben Gurion University of the Negev P. Figuerola received a fellowship from the FOMEC (University of Buenos Aires, Argentina) and from the CONICET (National Council for Science and Technology, Argentina).

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